Lecture 31: Diode Applications: Wave Shaping Circuits

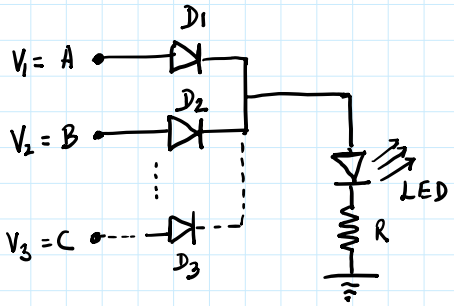
Logic Circuits Using Diodes: 4 important logic operations are:

1. OR
2. NOR
3. AND
4. NAND

(A) Logic OR operation: $C = A \oplus B$

Representation:



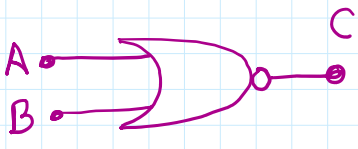


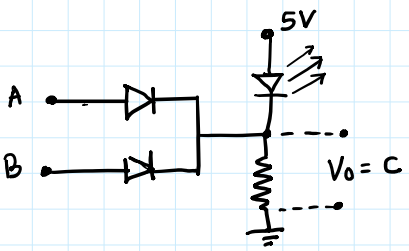
Truth Table: OR

A	B	C
0	0	0
0	1	1
1	0	1
1	1	1

(B) Logic NOR: $C = \overline{A \oplus B}$

Representation:





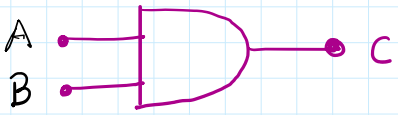
NOR logic gate

Truth Table: NOR

A	B	C
0	0	1
0	1	0
1	0	0
1	1	0

© Logic AND operation: $C = A \odot B$

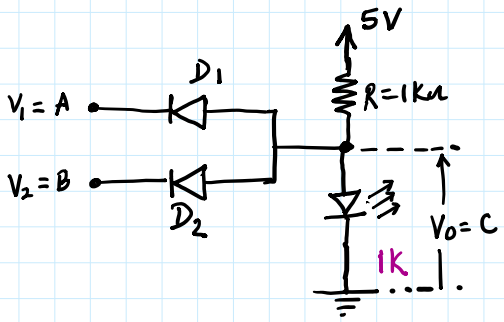
Representation:



$C = A \odot B$

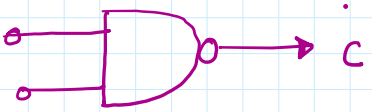
Truth Table: AND

A	B	C
0	0	0
0	1	0
1	0	0
1	1	1



Ⓓ Logic NAND operation: $C = \overline{A \odot B}$

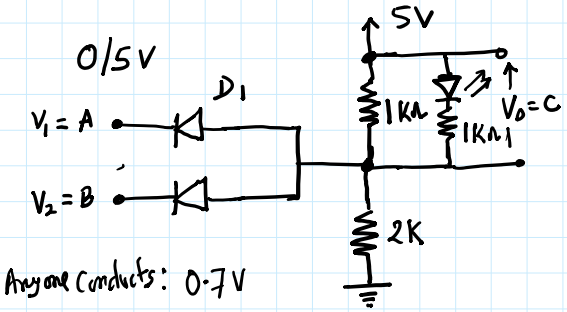
Representation:



$C = \overline{A \odot B}$

Truth Table: NAND

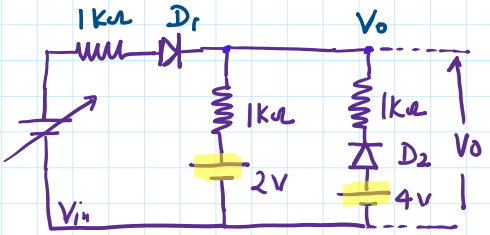
A	B	C	
0	0	1	
0	1	1	
1	0	1	
1	1	0	OFF



Example 2:

Q2 Compute Vo for

a) $V_{in} = 0V$
b) $V_{in} = 5V$
c) $V_{in} = 10V$



b) $V_{in} = 5V$

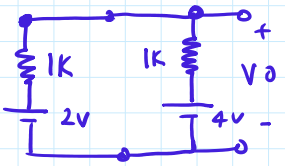
c) $V_{in} = 10V$



Solution:

a) $V_{in} = 0$

Let us assume, D_1 is OFF
D_2 is ON



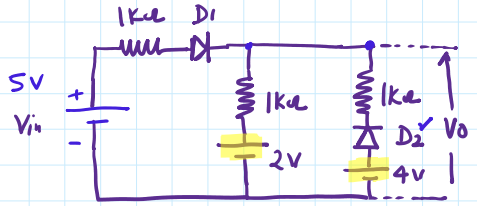
$$\frac{V_o - 2}{1} + \frac{V_o - 4}{1} = 0$$

$$\Rightarrow 2V_o = 6 \Rightarrow V_o = 3\ V$$

$V_o = 3V$ indicates that initial guess is correct as for $V_o = 3V$, D_1 is reversed biased and D_2 is forward biased.

b) When $V_{in} = 5V$

Assume { D_1 is ON, D_2 is ON



$$\frac{V_o - 5}{1} + \frac{V_o - 2}{1} + \frac{V_o - 4}{1} = 0$$

$$3V_o = 11,\ V_o = 11/3 = 3.6667$$

$V_o = 3.667\ V$ implies that D_1 forward biased and D is reversed biased. Hence initial guess is reasonable. In case...

Consider { D_1 ON, D_2 is OFF }

$$\frac{V_o - 5}{1} + \frac{V_o - 2}{1} = 0$$

$$V_o = \frac{7}{2} = 3.5\ V$$

$V_o = 3.5V < 4V$
thus D_2 will conduct. Hence the assumption D_2 is OFF is incorrect.

(c) $V_{in} = 10V$, assumption, D_1 ON, D_2 OFF is more reasonable.

For this $\frac{V_o - 10}{1} + \frac{V_o - 2}{1} + 0 = 0$

$V_o = 6\ V$ meets the criteria

D_1 ON and D_2 OFF.

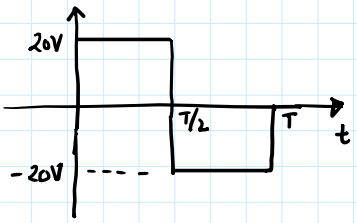
D_1 ON and D_2 OFF.

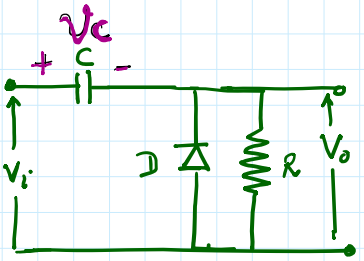
Example 3 : Problem 6 of Tutorial 5

For this given circuit, Sketch V_C and V_o.
Assume close to ideal diode, i.e., r_d is quite small.

(A) Assume, $RC = \tau \ll T/2$. and $r_d C = \tau' \ll \tau$
Where r_d = resistance of diode in on. state.

(B) Assume $RC = \tau \approx T/2$. Sketch V_C and V_o

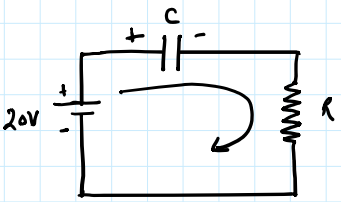


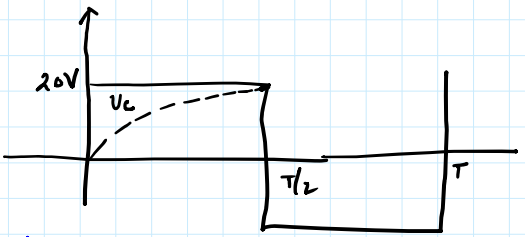


When 20V is applied at the input for a period of T/2, the diode is reverse biased, as shown in the figure.
The capacitor starts charging, through R.

with $V_{ini} = 0$, $V_f = 20V$,

$$V_C = V_f + (V_{ini} - V_f)\, e^{-t/RC}$$
$$= 20 + (0 - 20)\, e^{-t/RC}$$
$$= 20\,(1 - e^{-t/RC})$$
$$= 20\,(1 - e^{-t/\tau}); \quad \tau = RC$$



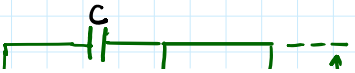


Now assume that time constant of RC circuit, $\tau = RC \ll T/2$
then V_c quickly reaches close to 20V.

The output voltage: $-V_i + V_C + V_o = 0$

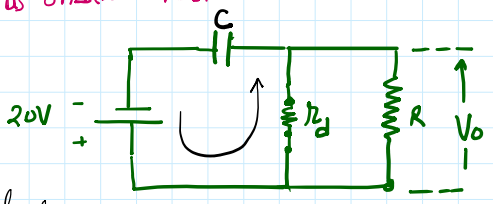
$$V_o = V_i - V_C$$
$$= 20 - 20\,(1 - e^{-t/\tau})$$
$$\boxed{V_o = 20\, e^{-t/\tau}} \quad ; \text{for } 0 < t \le T/2$$

Now when supply voltage changes to -20 V at $t = \frac{T}{2}^+$
Then diode is forward biased, with r_d as onstate resistance.
r_d is quite small hence relative



Then diode is forward biased; with r_d as onstate resistance.

r_d is quite small, hence capacitor first discharges to 0 and charges fast to −20 as $r_d * C = \tau'$ is very less.



V_c : $\underbrace{20}_{V_i} \rightarrow \underbrace{-20\,V}_{V_f}$

$$V_f + (V_i - V_f)\, e^{-t/\tau}$$

$$-20 + \{20 - (-20)\} e^{-t/\tau} = -20 + \{20 + 20\} e^{-t/\tau'}$$

$$= -20\,(1 - 2e^{-t/\tau'})$$

$$V_c = -20(1 - 2e^{-t/\tau'})$$

$$V_o = V_i - V_c = -20 - \{-20(1 - 2e^{-t/\tau'})\}$$

$$= -20 + 20\,(1 - 2e^{-t/\tau'}) = -20 + 20 - 40e^{-t/\tau'}$$

$$= -40e^{-t/\tau'}$$

$$\boxed{V_o = -40e^{-t/\tau'}} \quad \text{for } T/2 < t \le T$$

Next cycle when +20V is applied, initial voltage is −20V. Final voltage is +20V. Thus

$$V_c = V_f + (V_{ini} - V_f)\, e^{-t/\tau}$$

$$= 20 + (-20 - 20)\, e^{-t/\tau}$$

$$V_c = 20\,(1 - 2e^{-t/\tau}) \; ; \; T < t \le 3T/2$$

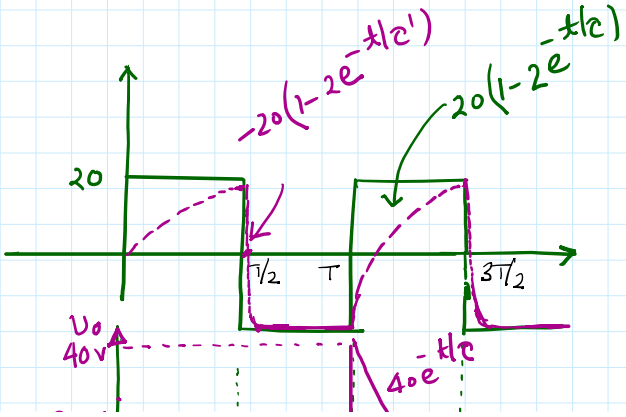
$$V_o = V_i - V_c = 20 - 20(1 - e^{-t/\tau}) = 40e^{-t/\tau}$$

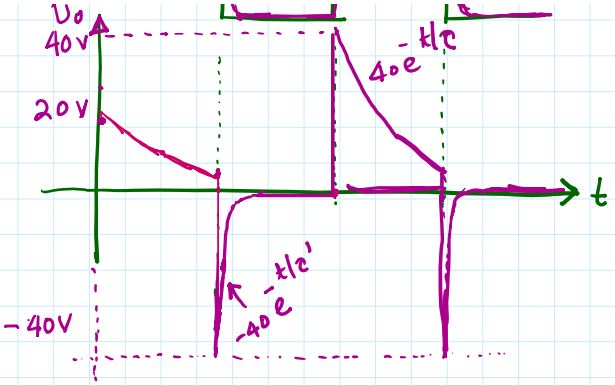
$$V_c = -20\,(1 - e^{-t/\tau'}) \; ; \; \frac{3T}{2} < T \le 2T \ldots$$

$$V_o = V_i - V_c = -20 - \{-20(1 - 2e^{-t/\tau'})\}$$

$$= -20 + 20\,(1 - 2e^{-t/\tau'}) = -40e^{-t/\tau'}$$

$\tau = RC$, $\tau' = r_d C$ for $\tau' << \tau << T/2$





(B) When $RC = \tau \approx T/2$, then during +20V input

$V_c = 20(1-2e^{-t/\tau})$ but τ is quite comparable with period T/2, it will reach to 20V fully, as shown below.

